

Pair-Conditioned Verification of Literature-Mined Taxa Interactions

Anonymous ACL submission

Abstract

Taxa interaction databases (who eats, infects or pollinates whom) are partly built by mining the literature. Candidates are proposed when a passage names two taxa near an interaction term, and the obvious filter asks whether the passage describes an interaction. A passage naming several taxa may support only some of its pairs, yet this filter gives them all the same answer. In our benchmark, passages naming three or more taxa hold 62% of the interacting pairs. Keeping a 110M-parameter encoder and its LLM-labelled training data fixed, we vary the input: the passage alone, the passage with the candidate pair, or the passage with the pair and its interaction term. On 437 expert-graded candidates, adding the pair raises the area under the precision-recall curve (AUPRC) from 0.867 to 0.942, and the interaction term adds no significant further gain. The gain is largest on passages naming three or more taxa. It replicates on the public BioRED corpus and in zero-shot Qwen models from 1.7B to 122B parameters, whose answers improve with size much more when asked about the pair. Asked only about the passage, the 32B model that labelled our training data ranks candidates below the 110M encoder given the pair.

1 Introduction

Databases such as GloBI (Poelen et al., 2014) record which taxa eat, infect, parasitise or pollinate which, and like many biomedical databases (Hirschman et al., 2012; Lee et al., 2018) they are populated in part by text mining (Dimitrova et al., 2020). A retrieval system proposes candidate triples (s_1, r, s_2) , two organisms and an interaction term, together with the passage that allegedly supports them, and something must decide which candidates a curator should review. A natural choice is a sentence-level classifier that reads the passage and decides whether it describes a biotic interaction.

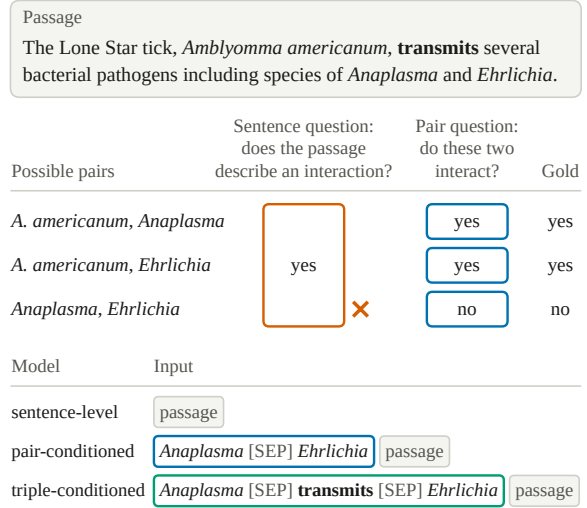


Figure 1: One passage, several candidate pairs. Retrieval proposes a candidate when a passage names two organisms near an interaction word (bold). The sentence question gets one answer for the whole passage, so even a perfect answer accepts the third pair, two bacterial genera merely listed together. The third pair is a real benchmark item (TEST299, graded negative) that the sentence-level model accepts and both conditioned models reject at their block-held-out thresholds; the first two pairs are illustrative. Bottom: model inputs.

This classifier gives one answer per passage, whereas co-occurrence retrieval proposes candidates pair by pair: a passage naming two taxa near an interaction term yields a candidate whether or not it asserts that interaction between those two. Every candidate therefore already contains the wording a sentence-level classifier responds to: in the retrieval pool we study (§3.1), all 175,588 candidates carry an interaction term, and a lexical scan for GloBI interaction vocabulary fires on 99.83% of their passages. The decision that matters is narrower: does the passage assert an interaction between these two taxa? A passage that names several organisms can support one pair and not another, and a classifier that sees only the passage gives the

same answer whichever pair the candidate names; even a perfect answer to its question accepts the unsupported pair (Figure 1).

Ruch et al. (2024), who built the retrieval system, graded 100 random BiotXplorer triples, found that the system identified the interacting taxa with a precision of 31%, and named the main cause of error: “instances where passages listed multiple species, which can be automatically filtered out.” On our benchmark (TAXAPAIRS, §3.3), passages naming three or more taxa hold 156 of the 251 interacting pairs (62%), so filtering them out would discard most of what the pipeline exists to find. We keep those passages and judge each candidate by its pair.

To test this, we train the same 110M-parameter encoder three times on the same corpus, labelled by a 32B language model, reading the passage alone (*sentence-level*), the passage with the candidate pair (*pair-conditioned*), or the passage with the whole candidate triple (*triple-conditioned*), and compare them on 437 expert-graded candidates. Everything except the input is held fixed. Telling the model which pair it is judging raises the area under the precision-recall curve (AUPRC, which summarises precision over all recall levels; a random ranking scores the share of positives) from 0.867 to 0.942 and F1 from 0.790 to 0.882 (McNemar $p = 3.8 \times 10^{-7}$), and at equal review workload keeps F1 higher at every share of candidates accepted from 4% to 89% (Figure 2b); adding the relation term gives no significant further gain (§4.1). On BioRED (Luo et al., 2022), a public biomedical corpus, the same comparison raises AUPRC from 0.738 to 0.843 and entity-pair F1 from 57.5 to 67.1, with the gain where the argument predicts: absent on sentences that name only two entities, large on sentences that name more. Zero-shot Qwen models from 1.7B to 122B parameters show the same gap, and the verifier runs at 32 candidates per second on a CPU.

Contributions.

- A controlled comparison of sentence-level, pair-conditioned and triple-conditioned verification on one encoder and one corpus, replicated on a public benchmark and in zero-shot Qwen language models from 1.7B to 122B parameters, with the gain located where the argument predicts (§4).
- An account of what the pair query contributes, namely disambiguation between the organ-

isms a passage names and a check on candidates whose arguments are not organisms, and a bound showing that checking a triple’s components separately cannot reach the largest error class (§4.3, §4.6).

We also give operating curves for choosing a threshold and a practical configuration with candidate rules, a four-way direction label, CPU throughput and escalation to a zero-shot LLM (§5; Appendices L–N), and six negative results (Appendix P).¹

2 Related Work

Partial-input baselines. That a classifier can score well by exploiting how its data were assembled is well established in NLP. Partial-input baselines recover much of the label from the hypothesis alone in natural language inference (Gururangan et al., 2018; Poliak et al., 2018) and from the passage alone in reading comprehension (Kaushik and Lipton, 2018), and such rules hold in distribution while failing under deployment (Geirhos et al., 2020; Du et al., 2024); the closest task-level precedent is fact verification (Thorne et al., 2018), where a claim-only classifier matches evidence-aware models because of how the corpus was built (Schuster et al., 2019). The sentence-level filter of §1 is a partial-input model in exactly this sense, except that the missing input is the query the pipeline itself issued rather than an annotator artefact (Wiegand et al., 2019; Parmar et al., 2023). As Feng et al. (2019) caution, our claims are limited to the models and corpora we study.

Entity pairs in relation extraction. Distant supervision assumes that a sentence containing two related entities expresses the relation (Mintz et al., 2009), and its standard relaxation exists because it does not (Riedel et al., 2010). Since TACRED an instance has been the triple (sentence, e_1 , e_2) rather than the sentence (Zhang et al., 2017), and how the pair is injected into the encoder changes F1 by several points (Baldini Soares et al., 2019; Wu and He, 2019; Zhou and Chen, 2022). Rosenman et al. (2020) already name our failure, as an “event heuristic” that classifies on (sentence, relation) while ignoring the arguments. Our contribution is a controlled measurement of its effect in a verification setting built by co-occurrence retrieval, replicated on BioRED and in zero-shot Qwen models and located where the argument predicts, and

¹Code: <https://anonymous.4open.science/r/XXXX>.

a bound on component-wise remedies. The entity-type shortcut usually invoked for such errors (Peng et al., 2020; Mtumbuka and Schockaert, 2024) is unavailable here, since both slots are filled from one inventory of taxon surface forms. Error attribution across pipeline stages is standard and easy to get wrong (Taillé et al., 2020; Ai and Kavuluru, 2023).

Conditioning on the query. Casting a relation as a question about a named entity (Levy et al., 2017), or as an entailment hypothesis verbalising the relation between two of them (Obamuyide and Vlachos, 2018; Sainz et al., 2021), is the reformulation our input format instantiates, and the resulting decision (does this passage support this specific claim?) has the shape of scientific claim verification (Wadden et al., 2020), and of validating a knowledge-graph triple against text (Lehmann et al., 2012), except that those systems retrieve evidence for a given triple, while here retrieval fixes the passage. Distilling a large model’s judgments into a small student is standard (West et al., 2022; Gekhman et al., 2023). Selective classification thresholds a score instead of issuing a verdict (Geifman and El-Yaniv, 2017), which is how extraction is made useful to human abstractors (Swaminathan et al., 2024).

Text mining for interaction databases. Judging an extracted fact before it reaches a curator is an old idea (Rodriguez-Esteban et al., 2006), as is evaluating such tools by whether they help the curator (Hirschman et al., 2012; Britan et al., 2018; Lee et al., 2018). Cuzick et al. (2023), independently, take the interacting pair rather than the passage as the annotation unit, the unit our pair-binding error class (§4.6) isolates. Gold corpora for biodiversity text annotate entities and the relations between them, such as microorganisms with their habitats and phenotypes (Bossy et al., 2019) or organisms with environments, processes and qualities (Abdelmageed et al., 2022); they serve to train and score extraction from text, whereas our task is to verify candidates that retrieval has already proposed. Large language models mine ecological text (Farrell et al., 2024; D’Souza et al., 2025) and extract interactions at scale (Keck et al., 2025; Zou et al., 2026); Zou et al. (2026) standardise names and categories post hoc, with no step that checks each extraction against its text. We are not aware of published work that verifies literature-mined biotic-interaction triples against their source passage before ingestion.

3 Method

3.1 Candidates and task definition

BiotXplorer (Ruch et al., 2024) proposes candidate biotic-interaction triples from the literature indexed by SIB Literature Services (Gobeill et al., 2020). For each passage it matches taxon surface forms against a taxonomy and interaction surface forms against a vocabulary derived from the Relation Ontology, and emits candidate (s_1, r, s_2) triples with the passage, the matched surface strings and their canonical forms. We use it only as a source of candidates: no component of it is a baseline in this paper. The pool we study has 175,588 candidate rows over 165,325 distinct passages and 115,404 distinct triple keys. Canonical strings often differ from what the passage says, so we query with the surface form, which is what the model can see. The decision to make is whether the passage asserts an interaction between the candidate’s two organisms, in either direction (§3.5).

3.2 Training data and models

The teacher. Labels come from a locally hosted 32B open-weight instruction model (Qwen Team, 2025), prompted zero-shot at temperature 0 (prompts in Appendix A) and asked, per candidate, whether the passage supports this triple. Its labels therefore vary across candidates that share a passage. Against 150 expert-graded items its mislabel rate is $22/150 = 14.7\%$ (Clopper–Pearson [9.4%, 21.4%]; 10 false positives, 12 false negatives).

The corpus. 48,338 teacher-labelled rows, 0.343 positive, drawn from the same retrieval pool as the benchmark (disjoint passages): a 20,000-row base sample, 16,546 *contrast* rows, in which one passage and its pair recur with different relation terms (on 2,017 passages with labels that disagree), 6,201 actively sampled rows, and 5,591 rows with the candidate’s argument order reversed.

Models and inputs. All three are cross-encoders on BiomedBERT-base (Gu et al., 2021) (110M parameters), distilled from the teacher’s labels (Bucilua et al., 2006; Hinton et al., 2015) with the same recipe (Appendix O), keeping the checkpoint with the best development AUPRC on a 10% development split grouped by taxon pair so that no pair spans training and development. Three seeds each; one run takes about four minutes on a single A100. They differ only in their input. The **sentence-level**

model reads the passage alone. It is the simplest formulation, and on rows that share a passage it is shown the same input with different labels, which it cannot fit. The **pair-conditioned** model reads the two candidate taxa, s_1 [SEP] s_2 , as segment A and the passage as segment B (Nogueira and Cho, 2019). The **triple-conditioned** model reads the whole candidate, s_1 [SEP] r [SEP] s_2 . Comparing the three separates the effect of the pair from that of the relation term.

3.3 An expert-graded benchmark

Our benchmark, TAXAPAIRS, has 437 rows, 251 positive, in three blocks of different provenance; we report every block as well as the pooled figures, since pooling alone hides their disagreement (§4.3, Appendix C). BIOTX100 consists of retrieved candidates, each with one passage chosen by retrieval, expert-graded on four axes: whether each taxon surface form was correctly resolved, whether the relation was correct, and whether the whole triple is supported by its passage. Half were drawn from pairs already recorded in GloBI and half at random, and after removing three training duplicates it contributes 97 rows: 50 GloBI-matched with 43 positive, 47 random with 33 positive. REJECT50 consists of candidates an earlier filter rejected, expert-graded for the ones it should have kept: 41 clean rows, 11 positive. It is the paper’s recall-side instrument (Appendix I). TEST299 is 299 expert-graded pair-level items, 0.548 positive, and is where passages most often name several taxa.

The blocks do not carry the same gold: BIOTX100 alone is graded at the level of the whole triple, while TEST299 was expert-graded at the pair level and its stored relation term was attached by retrieval. We therefore evaluate everywhere at the level all three share (“do these two taxa interact in this passage?”), and use the triple-level grades only to diagnose (§4.6). This can understate any benefit of triple conditioning, since a candidate with the right pair and the wrong relation counts as correct.

One caveat applies: we scored roughly a hundred model configurations on this benchmark during development, so the recipes compared here were chosen with it in view; we expect the controlled comparison of §4 to be the least affected.

3.4 Replication setting: BioRED

TAXAPAIRS is ours, built and graded by three of the authors. To test the claim where anyone can check it, we rebuild the same setting on BioRED

(Luo et al., 2022), a public corpus of PubMed abstracts with expert-annotated entities and relations. Candidates are generated by co-occurrence: every pair of distinct annotated concepts co-mentioned in one sentence, restricted to the eight entity-type pairs that carry relations in training, with the sentence as passage. A candidate is positive when BioRED relates the two concepts anywhere in the abstract, a document-level label whose noise every model shares. We follow the protocol of Lai et al. (2025): models train on the 500 abstracts of BioRED’s training and development splits (15,640 candidates), select their epoch and threshold on its 100 test abstracts, and are scored on the 400 abstracts of the BioCreative VIII test set (Islamaj et al., 2024). BioRED has no retrieved relation term, so only the sentence-level and pair-conditioned formulations are compared, trained with the recipe of §3.2.

3.5 Evaluation protocol

The threshold policy is fixed in advance. A model’s reported operating point is either **pre-specified**, chosen on its own training-development split and never on the benchmark, or **block-held-out**: fitted on two of the three source blocks and applied to the third, so that every row is scored by a threshold that never saw it and every row is used. Comparisons between models use the same policy for both. Intervals are Clopper–Pearson; McNemar tests (McNemar, 1947) carry a continuity correction. With three seeds we report either the per-seed mean and standard deviation or a three-checkpoint ensemble, which averages the checkpoints’ probabilities. The benchmark ships an `in_train` flag on 9 rows; a 5-gram Jaccard scan against all 48,338 training rows finds 3 more at Jaccard 1.000, all in BIOTX100. All 12 are excluded throughout.

The gold is pair-level and therefore order-free, so a model that reads the candidate in order is scored as the higher of its two scores over the two argument orders, that is, the probability that the pair interacts in at least one direction. Otherwise a model would be penalised for rejecting a correctly paired candidate whose stated relation runs backwards, which the gold does not count as an error.

4 Results

4.1 Main results

Table 1 gives the comparison. On TAXAPAIRS, telling the model which pair a candidate is about

Model	TAXAPAIRS (437)		BioRED (16,606)		
	AUPRC	F1	AUPRC	F1	EP F1
Accept all	0.574	0.730	0.463	0.633	–
Sentence-level	0.867 $\pm$.006	0.790	0.738 $\pm$.004	0.718	57.5
Pair-conditioned	0.942 $\pm$.007	0.882[†]	0.843$\pm$.004	0.817[†]	67.1
Triple-conditioned	0.945$\pm$.005	0.867 [†]	–	–	–

Table 1: Within each benchmark, all models share one encoder and one training corpus; only the input differs. AUPRC: mean $\pm$ s.d. over three seeds. F1: three-checkpoint ensemble at block-held-out thresholds (TAXAPAIRS) or the pre-specified threshold (BioRED). EP F1: BioRED’s document-level entity-pair F1 (%), seed mean. Accept all gives the base rate. [†]McNemar $p < 10^{-4}$ against the sentence-level model; pair-against triple-conditioned, $p = 0.26$. BioRED has no relation term, hence no triple-conditioned model. Per-seed values (TAXAPAIRS): Appendix B.

raises per-seed AUPRC from 0.867 ± 0.006 to 0.942 ± 0.007 and, at block-held-out thresholds, F1 from 0.790 to 0.882 (precision 0.752 to 0.869, recall 0.833 to 0.896). On decision correctness the pair-conditioned model fixes 74 items and breaks 23, McNemar $p = 3.8 \times 10^{-7}$. The advantage does not depend on calibration: compared at equal review workload, the pair-conditioned model is ahead by 0.07 to 0.09 F1 whenever 50% to 70% of candidates are accepted, and the curves come within 0.01 F1 of each other only when fewer than 10% or more than 85% of candidates are accepted (Figure 2b; shares at six thresholds in Appendix C).

Adding the relation term gives no significant further gain. The triple-conditioned model’s AUPRC is level with the pair-conditioned model’s (0.945 against 0.942), and on decision correctness it fixes 21 items and breaks 30 ($p = 0.26$).

4.2 Results on BioRED

The result (Table 1, right; full table in Appendix D) reproduces the one on TAXAPAIRS. Conditioning on the pair raises AUPRC from 0.738 to 0.843 and F1 from 0.718 to 0.817, fixing 2,642 items and breaking 1,064. Marking the two mentions in the passage, the standard relation extraction input (Zhou and Chen, 2022), adds nothing to the pair query ($p = 0.08$).

On BioRED’s standard document-level metric the pair-conditioned model reaches an entity-pair F1 of 67.1, against 57.5 for the sentence-level one. Systems that read the whole abstract reach 74.1 to 75.3 (PubMedBERT, BioREx and BioREDirect; Lai et al., 2023, 2025) and zero-shot GPT-4 42.9 (Islamaj et al., 2024). Much of that gap is

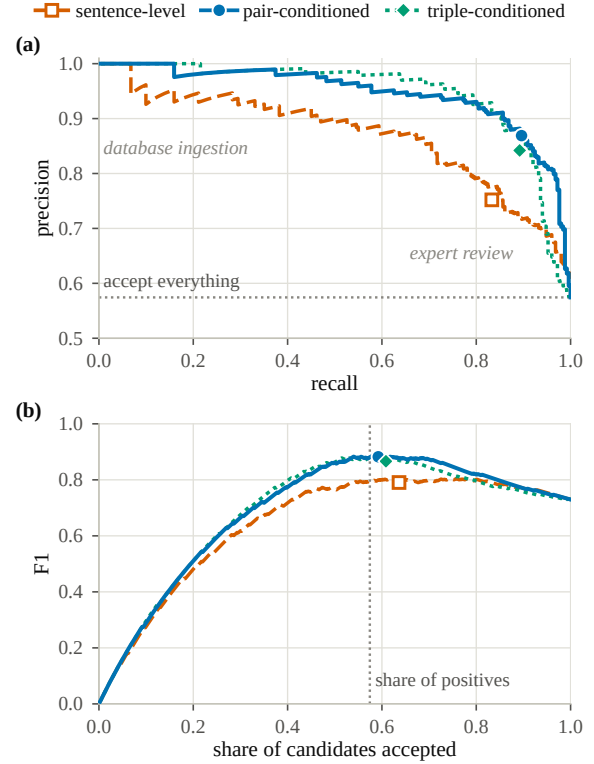


Figure 2: Operating curves on the 437 expert-graded candidates of TAXAPAIRS (three-checkpoint ensembles). (a) Precision against recall. Markers are the operating points of Table 1; their thresholds are fitted per block, so they can sit slightly off the pooled curve. Dotted line: accepting every candidate. (b) F1 against the share of candidates accepted, which compares the models at equal review workload whatever their calibration; the pair-conditioned model is ahead of or level with the sentence-level one at every share except 93% to 96%, where it trails by at most 0.003. Values at six thresholds: Appendix C.

structural: a quarter of the related pairs are never co-mentioned in one sentence, which caps any sentence-scope verifier at recall 0.749. Given the whole abstract, the same model and recipe reach 67.0 (recall 0.697, precision 0.646). The published systems also classify entity pairs, so the remaining gap comes from their document-level modelling, which is a separate problem from ours.

4.3 Gain by number of taxa in the passage

Figure 3 locates the gain; it is the most direct test of the argument. On passages that name three or more taxa, adding the pair raises AUPRC by 0.098; on passages naming two or fewer taxa, by 0.043. Splitting by argument shows where this smaller gain comes from: the candidates come from automatic matching, and 41% of the 437 have one

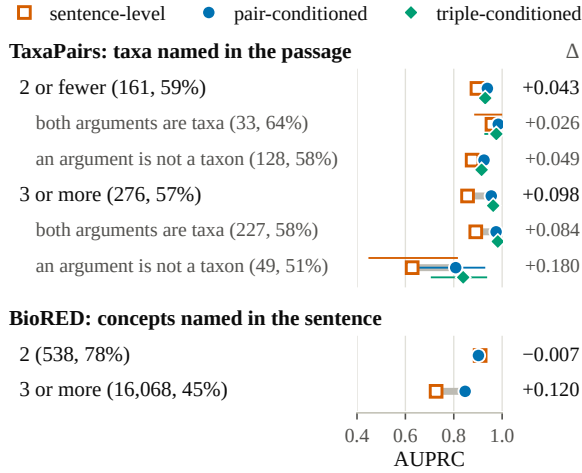


Figure 3: AUPRC by the number of entities the passage names (three-checkpoint ensembles). TAXAPAIRS: distinct taxa that TaxoNERD (Le Guillaume and Thuiller, 2022) finds in the passage; an argument “is not a taxon” when TaxoNERD does not recognise it there. BioRED: gold-annotated concepts in the sentence; BioRED has no relation term, so there is no triple-conditioned model. Triple-conditioned markers sit just below their row so they do not hide the pair-conditioned ones. Grey bars and the Δ column give the gain from adding the pair (pair-conditioned minus sentence-level). Parentheses give the number of candidates and the share of positives. Thin lines on the rows under 100 candidates are bootstrap 95% intervals (2,000 resamples of candidates, fixed seed; values released with the code). Exact values: Appendices C and D.

“taxon” that is a string TaxoNERD does not recognise as an organism in the passage (an organ, an author’s name, a vernacular term). Where both arguments are recognised taxa and the passage names no third one, the case in which the sentence question and the pair question nearly coincide, the pair query adds +0.026 on 33 items; where an argument is not a taxon it adds +0.049, and +0.180 when the passage also names others. The pair query therefore does two things. It tells the model which pair to judge, and it lets the model check that both arguments are organisms. BioRED, whose entities are gold-annotated, isolates the first: on candidates from sentences that name only two concepts the two models are indistinguishable (0.909 against 0.902; per seed 0.899–0.919 against 0.892–0.914), while on those from sentences naming more, which are 97% of the candidates, conditioning adds +0.120. By source block, the pair query raises AUPRC from 0.835 to 0.944 on TEST299, whose passages name the most taxa, from 0.964 to 0.982 on BIOTX100, and from 0.551 to 0.575

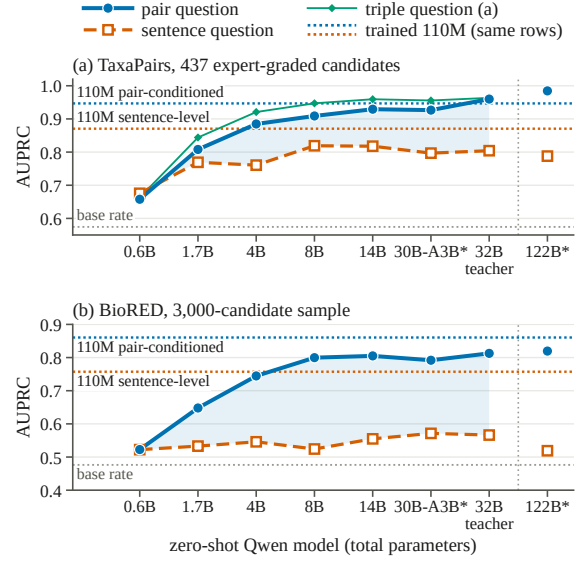


Figure 4: Zero-shot Qwen models (Qwen Team, 2025) asked the sentence question (dashed, open squares) or the pair question (solid, filled circles) on (a) the 437 expert-graded candidates of TAXAPAIRS and (b) a 3,000-candidate BioRED sample. Dotted: the trained 110M models on the same rows (three-checkpoint ensembles); grey: base rate. The x-axis spaces model sizes evenly. From 1.7B on, the pair question leads at every size, while answers to the sentence question change little with scale. *Mixture of experts, placed by total size; the 32B model is the teacher. Qwen3.5-122B-A10B is a later release. Thin green line in (a): the triple question (TAXAPAIRS only), which matches or exceeds the pair question (Appendix E).

on the 41-row REJECT50, too small to rank the models (Appendices C and I).

4.4 Zero-shot language models

The formulation effect is not a property of small encoders (Figure 4; scores are $P(\text{YES})$, Appendix A). Asked the sentence question, the 32B teacher scores AUPRC 0.804, below the 110M sentence-level student; asked about the pair, 0.960. Its gap, +0.156, is twice the student’s (the two questions also differ in their instructions; Limitations), and it sits in the same place: +0.179 on passages naming three or more taxa against +0.107 on the rest, where the student’s is +0.098 against +0.043. The 110M pair-conditioned student, about 300 times smaller than its teacher, scores 0.14 AUPRC higher than the teacher asked the sentence question. As with the student, the relation term adds nothing at this size (0.964 against 0.960), although zero-shot models from 1.7B to 30B-A3B do gain from it (Appendix E). From Qwen3-1.7B to Qwen3.5-

122B-A10B (Qwen Team, 2026) the pair question leads at each size, by +0.04 to +0.20 AUPRC on TAXAPAIRS and +0.11 to +0.30 on BioRED, while the sentence question gains little with size. On BioRED the teacher, asked the sentence question, accepts 92% of its sample (AUPRC 0.566 at a base rate of 0.48); asked about the pair, 0.813, still behind the trained pair-conditioned model (0.860 on the same sample). Against its teacher’s labelling prompt (0.964) the pair-conditioned student gives up 0.021 AUPRC, of which distilling the teacher’s probabilities recovers about a quarter (Appendix F). In F1 it gives up almost nothing at block-held-out thresholds (0.882 against the teacher’s 0.885; the soft-label student reaches 0.883).

4.5 Controls: query ablation, training data, encoder

Three controls locate the effect in the query. Degrading the trained triple-conditioned model’s query at inference, weights and thresholds unchanged, shows which part of the query it uses: deleting both taxa drops the three-checkpoint ensemble’s AUPRC from 0.948 to 0.691 against a base rate of 0.574, and emptying the query gives 0.649, while replacing or shuffling the relation, pair kept, moves it far less (0.906, 0.930; Appendix G). The conditioned model does not need the corpus’s contrast rows: a triple-conditioned model trained on the 20,000-row base sample alone reaches AUPRC 0.943 ± 0.004 (full corpus 0.945 ± 0.005) and block-held-out F1 0.886. The encoder is not the cause either: of seven alternative recipes screened on development data, the three that passed were scored once on test, none gains significantly on both benchmarks, and with every encoder trained in both formulations the pair query wins, by +0.103 to +0.113 AUPRC on BioRED and +0.051 to +0.076 on TAXAPAIRS (Appendix H).

4.6 Error analysis

The pair-conditioned model’s 34 false positives at its block-held-out operating point are not homogeneous: 24 come from passages naming three or more taxa, and 16 have an argument TaxoNERD does not recognise as a taxon. Read by hand, they sort into a few mechanisms (examples in Table 10, Appendix J; the per-row table is released): *pair-binding*, where both taxa and the relation are correct but the relation’s argument is a third organism; *co-listing*, where the two taxa are members

of one enumeration and the passage states no interaction between them; *negation*, where the passage asserts the opposite; *same organism*, where the two “taxa” are one organism named twice or an organism and its own clade; candidates whose relation term names no biotic interaction at all (*relative to*, *hybridization*, *migration*); and *entity* failures, where a taxon string is a taxonomic author, a place, an organ, or an adjective inside a pathogen’s name (“avian schistosomes”, which the pair-conditioned model already rejects).

A bound on component-wise verification. The pair-binding class limits a natural alternative architecture. Its members have both taxa correctly resolved and the correct relation; the error is a pairing made by retrieval that the passage does not state. A verifier that checks s_1 , r and s_2 separately, however good each check is, cannot see it. On the 97 clean BIOTX100 candidates, where triple-level grades make the classes countable, 15 of the 32 unsupported candidates are pair-binding failures (the rest: 9 entity, 8 predicate), a share of 0.47, Clopper–Pearson [0.29, 0.65]. A component-wise verifier accepting a fraction r of the 65 supported triples therefore also accepts all 15, so its precision is at most $65r/(65r + 15)$: 0.81 at full recall, [0.76, 0.88] across that interval. Only a model that reads the pair together with its passage can exceed this bound.

4.7 Fresh candidates

We also scored 100 candidates from outside TAXAPAIRS: two further batches of 50 from the same retrieval system, graded once at the pair level in the same expert curation as TEST299, 32 of them positive (a base rate of 0.32, against 0.574 on the benchmark). None of their passages occurs in the benchmark or in the training corpus. All three models score well below their benchmark figures, and the sentence-level model is again last: AUPRC 0.665 sentence-level, 0.785 pair-conditioned and 0.765 triple-conditioned (three-checkpoint ensembles). Unlike the benchmark’s candidates, these were scored with the retrieval system’s canonical names for the taxa and the relation, which for most of them differ from the passage’s wording (§3.1), so the two conditioned models’ figures are not directly comparable with their benchmark figures; the sentence-level model reads no query and is unaffected. With 32 positives the size of the drop is uncertain, but it suggests that the benchmark is

easier than a live pool, so its absolute figures are likely to be optimistic.

5 Discussion and Conclusion

Partial input and model size. A sentence-level filter is a partial-input model in the sense of §2, except that the input it lacks is the query the pipeline issued, so the remedy is to put that query into the input. Larger models do not make up for the missing input. From 1.7B to 122B, zero-shot Qwen models answer the pair question better at each size while their answers to the sentence question gain little, and a 110M model asked about the pair ranks candidates better than the 32B teacher asked about the passage (§4.4).

Passages naming only two taxa. The two questions coincide when a passage names only the candidate’s two entities and both are what they claim to be: on BioRED sentences that name two concepts the two models are indistinguishable, and on TAXAPAIRS the gain is smallest where both arguments are recognised taxa and no third is named (§4.3). Most of its positives come from passages that name more.

Choosing an operating point. The precision of the pair-conditioned model ranges from 0.761 at $\tau = 0.01$ to 0.981 at $\tau = 0.90$ (Appendix C), so the right threshold depends on the use. Ingestion into a reference database calls for high precision, and assembling candidates for expert review calls for high recall. Reporting a score leaves this choice to the user (Geifman and El-Yaniv, 2017).

A practical configuration. For use in a pipeline we recommend a variant of the pair-conditioned model that adds deterministic candidate rules and a head labelling the interaction’s direction in four categories (Appendices L and M). Its input and training labels also differ from those of the pair-conditioned model, so it lies outside the controlled comparison. It scores 32 candidates per second on a CPU, and it can escalate the candidates it is least sure of to a zero-shot LLM (Appendix N).

Errors made before verification. Errors such as a wrong interaction term or a mis-resolved taxon arise during retrieval, before any verifier runs, and a model that reads the passage cannot see that the resolved taxon is not the one the passage names (Limitations). Direction needs its own gold (Appendix K).

Other domains. Any pipeline that proposes candidates by co-occurrence and filters them passage by passage gives every candidate from a passage the same answer; BioRED shows the remedy carrying over to genes, chemicals and diseases. For another pipeline, the first quantity to measure is the share of passages that name more than the candidate pair.

Conclusion. When candidates come from co-occurrence retrieval, the model that verifies them should see the candidate pair as well as the passage. A classifier that reads only the passage gives every pair in it the same answer, and 62% of the interacting pairs in TAXAPAIRS come from passages naming three or more taxa. Giving the model the pair raises AUPRC from 0.867 to 0.942, with the largest gain on those passages. The effect replicates on BioRED and in zero-shot Qwen models from 1.7B to 122B parameters. The verifier runs on a CPU, fast enough to score every candidate in the pool.

Limitations

Benchmark size and annotation. TAXAPAIRS has 437 expert-graded rows in three blocks, each graded once. Labels were adjudicated by a blind second annotation of 22 candidates: 11 on which two models (triple-conditioned and recommended, §5) confidently disagreed with the gold, and 11 random controls. It kept every control label and changed 7; on the earlier labels the pair query raises AUPRC from 0.851 to 0.918. On the recall side (REJECT50, Appendix I) the denominator is 11 positives.

Rule selection. Four of the eight candidate rules were selected after reading this benchmark’s false positives (Appendix L). The training-data criterion bounds how much that selection can have memorised the benchmark, but it does not remove it, and their effects should be read as optimistic until confirmed on candidates neither the rules nor their author have seen. These four reject at most one of the fresh candidates of §4.7, too few to confirm or contradict them.

Fresh candidates. Scores drop on them (§4.7).

Pair-level evaluation. Only BIOTX100 carries whole-triple gold. Evaluating at the pair level everywhere means a candidate with the right pair and a misbound relation counts as correct, so the

comparison cannot credit the relation term with the cases it alone could catch; measuring those would need triple-level gold on the other two blocks.

Document-level labels in BioRED. A candidate is positive when the abstract relates the two concepts anywhere, so a co-mention in a sentence that does not state the relation counts as positive. Both models are trained and scored on the same noisy labels, so the comparison is controlled, but the absolute figures understate what either would reach against sentence-level gold.

Prompt wording. Beyond naming the pair, the zero-shot pair question on TAXAPAIRS (Appendix A) asks the model to check that both strings name organisms in the passage and to reject a mere co-mention or an interaction each has with a third organism, and it omits the sentence question’s examples of interactions (on BioRED its only added instruction is the co-mention clause), so part of its lead in §4.4 may come from this wording rather than from naming the pair. The trained models read no instructions and learn from the same labels; their inputs differ only by the pair and, for the triple-conditioned model, the relation term, so the controlled comparison of Table 1 is unaffected.

Teacher noise and pretraining exposure. The training labels come from a prompted teacher wrong about one time in seven, and the student inherits that boundary; training on model-generated labels can degrade encoders (Lu and Smith, 2025). The passages are published MEDLINE text and the teacher is a pretrained model, so we cannot exclude that it saw them, and its labels may be optimistic for genuinely new literature.

Upstream failures. Of the 41 REJECT50 candidates, 27 carry an incorrect interaction term and 12 a mis-resolved taxon, 10 of them both. Those fail before any verifier is consulted, and no change to the decision stage, ours included, addresses them.

Single retrieval system and domain. We study one retrieval system with one interaction vocabulary, on interactions between taxa in MEDLINE. The mechanism, a pair-agnostic decision over passages that name several organisms, is not specific to this pool, but we have not shown that other pools fail the same way.

References

- Nora Abdelmageed, Felicitas Löffler, Leila Feddou, Alsayed Algergawy, Sheeba Samuel, Jitendra Gaikwad, Anahita Kazem, and Birgitta König-Ries. 2022. [Bio-divNERE: Gold standard corpora for named entity recognition and relation extraction in the biodiversity domain](#). *Biodiversity Data Journal*, 10:e89481.
- Xuguang Ai and Ramakanth Kavuluru. 2023. End-to-end models for chemical–protein interaction extraction: Better tokenization and span-based pipeline strategies. In *2023 IEEE 11th International Conference on Healthcare Informatics (ICHI)*, pages 610–618. IEEE. ArXiv:2304.01344.
- Livio Baldini Soares, Nicholas FitzGerald, Jeffrey Ling, and Tom Kwiatkowski. 2019. [Matching the blanks: Distributional similarity for relation learning](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 2895–2905, Florence, Italy. Association for Computational Linguistics.
- Robert Bossy, Louise Deléger, Estelle Chaix, Mouhamadou Ba, and Claire Nédellec. 2019. [Bacteria Biotope at BioNLP Open Shared Tasks 2019](#). In *Proceedings of the 5th Workshop on BioNLP Open Shared Tasks*, pages 121–131, Hong Kong, China. Association for Computational Linguistics.
- Aurore Britan, Isabelle Cusin, Valérie Hinard, Luc Motin, Emilie Pasche, Julien Gobeill, Valentine Rech de Laval, Anne Gleizes, Daniel Teixeira, Pierre-André Michel, Patrick Ruch, and Pascale Gaudet. 2018. [Accelerating annotation of articles via automated approaches: evaluation of the neXtA5 curation-support tool by neXtProt](#). *Database*, 2018:bay129.
- Cristian Buciluă, Rich Caruana, and Alexandru Niculescu-Mizil. 2006. Model compression. In *Proceedings of the 12th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 535–541.
- Alayne Cuzick, James Seager, Valerie Wood, Martin Urban, Kim Rutherford, and Kim E. Hammond-Kosack. 2023. [A framework for community curation of interspecies interactions literature](#). *eLife*, 12:e84658.
- Mariya Dimitrova, Jorrit Poelen, Georgi Zhelezov, Teodor Georgiev, Donat Agosti, and Lyubomir Penev. 2020. [Semantic publishing enables text mining of biotic interactions](#). *Biodiversity Information Science and Standards*, 4:e59036.
- Jennifer D’Souza, Zachary Laubach, Tarek Al Mustafa, Sina Zarriß, Robert Frühstückl, and Phyllis Illari. 2025. [Mining for species, locations, habitats, and ecosystems from scientific papers in invasion biology: A large-scale exploratory study with large language models](#). In *Proceedings of the 1st Workshop on Ecology, Environment, and Natural Language Processing (NLP4Ecology2025)*, pages 16–23, Tallinn, Estonia. University of Tartu Library.

746	Mengnan Du, Fengxiang He, Na Zou, Dacheng Tao, and Xia Hu. 2024. Shortcut learning of large language models in natural language understanding . <i>Communications of the ACM</i> , 67(1):110–120.	803
747		804
748		805
749		806
750	Maxwell J. Farrell, Nicolas Le Guillarme, Liam Brierley, Bronwen Hunter, Daan Scheepens, Anna Willoughby, Andrew Yates, and Nicole Mideo. 2024. The changing landscape of text mining: a review of approaches for ecology and evolution . <i>Proceedings of the Royal Society B: Biological Sciences</i> , 291(2027):20240423.	807
751		808
752		809
753		810
754		811
755		812
756	Shi Feng, Eric Wallace, and Jordan Boyd-Graber. 2019. Misleading failures of partial-input baselines . In <i>Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics</i> , pages 5533–5538, Florence, Italy. Association for Computational Linguistics.	813
757		814
758		
759		815
760		816
761		817
762	Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. In <i>Advances in Neural Information Processing Systems 30 (NIPS 2017)</i> , pages 4878–4887.	818
763		819
764		820
765		821
766	Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. Shortcut learning in deep neural networks . <i>Nature Machine Intelligence</i> , 2(11):665–673.	822
767		823
768		
769		824
770		825
771	Zorik Gekhman, Jonathan Herzig, Roei Aharoni, Chen Elkind, and Idan Szepkektor. 2023. TrueTeacher: Learning factual consistency evaluation with large language models . In <i>Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing</i> , pages 2053–2070, Singapore. Association for Computational Linguistics.	826
772		827
773		828
774		829
775		830
776		
777		831
778	Martin Gerner, Goran Nenadic, and Casey M. Bergman. 2010. LINNAEUS: A species name identification system for biomedical literature . <i>BMC Bioinformatics</i> , 11:85.	832
779		833
780		834
781		
782	Julien Gobeill, Déborah Caucheteur, Pierre-André Michel, Luc Mottin, Emilie Pasche, and Patrick Ruch. 2020. SIB literature services: RESTful customizable search engines in biomedical literature, enriched with automatically mapped biomedical concepts . <i>Nucleic Acids Research</i> , 48(W1):W12–W16.	835
783		836
784		837
785		838
786		839
787		
788	Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. 2021. Domain-specific language model pretraining for biomedical natural language processing. <i>ACM Transactions on Computing for Healthcare</i> , 3(1):1–23.	840
789		841
790		842
791		843
792		
793		844
794	Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. 2018. Annotation artifacts in natural language inference data . In <i>Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)</i> , pages 107–112, New Orleans, Louisiana. Association for Computational Linguistics.	845
795		846
796		847
797		848
798		
799		849
800		850
801		851
802		852
		853
		854
		855
		856
		857
		858
		859

860	volume 7649 of <i>Lecture Notes in Computer Science</i> ,	Adam Poliak, Jason Naradowsky, Aparajita Haldar,	914
861	pages 312–327. Springer.	Rachel Rudinger, and Benjamin Van Durme. 2018.	915
862	Omer Levy, Minjoon Seo, Eunsol Choi, and Luke	Hypothesis only baselines in natural language inference .	916
863	Zettlemoyer. 2017. Zero-shot relation extraction via	In <i>Proceedings of the Seventh Joint Confer-</i>	917
864	reading comprehension . In <i>Proceedings of the 21st</i>	<i>ence on Lexical and Computational Semantics</i> , pages	918
865	<i>Conference on Computational Natural Language</i>	180–191, New Orleans, Louisiana. Association for	919
866	<i>Learning (CoNLL 2017)</i> , pages 333–342, Vancouver,	Computational Linguistics.	920
867	Canada. Association for Computational Linguistics.		
868	Yucheng Lu and Kazimier Smith. 2025. Feeding LLM	Qwen Team. 2025. Qwen3 technical report. <i>arXiv</i>	921
869	annotations to BERT classifiers at your own risk.	<i>preprint arXiv:2505.09388</i> .	922
870	<i>arXiv preprint arXiv:2504.15432</i> .		
871	Ling Luo, Po-Ting Lai, Chih-Hsuan Wei, Cecilia N	Qwen Team. 2026. Qwen3.5: Towards native	923
872	Arighi, and Zhiyong Lu. 2022. BioRED: a rich	multimodal agents. https://qwen.ai/	924
873	biomedical relation extraction dataset. <i>Briefings in</i>	blog?id=qwen3.5 . Model card: https://	925
874	<i>Bioinformatics</i> , 23(5):bbac282.		926
875	Quinn McNemar. 1947. Note on the sampling error	Jonathan A. Rees and Karen Cranston. 2017. Auto-	928
876	of the difference between correlated proportions or	mated assembly of a reference taxonomy for phylo-	929
877	percentages. <i>Psychometrika</i> , 12(2):153–157.	genetic data synthesis . <i>Biodiversity Data Journal</i> ,	930
878	Mike Mintz, Steven Bills, Rion Snow, and Dan Juraf-	5:e12581.	931
879	sky. 2009. Distant supervision for relation extrac-	Sebastian Riedel, Limin Yao, and Andrew McCallum.	932
880	tion without labeled data. In <i>Proceedings of ACL-</i>	2010. Modeling relations and their mentions with-	933
881	<i>IJCNLP</i> , pages 1003–1011.	out labeled text . In <i>Machine Learning and Knowl-</i>	934
882	Frank Martin Mtumbuka and Steven Schockaert. 2024.	<i>edge Discovery in Databases (ECML PKDD 2010)</i> ,	935
883	Entity or relation embeddings? an analysis of encod-	Lecture Notes in Computer Science, pages 148–163,	936
884	ing strategies for relation extraction . In <i>Findings</i>	Barcelona, Spain. Springer Berlin Heidelberg.	937
885	<i>of the Association for Computational Linguistics:</i>	Raul Rodriguez-Esteban, Ivan Iossifov, and Andrey	938
886	<i>EMNLP 2024</i> , pages 6003–6022, Miami, Florida,	Rzhetsky. 2006. Imitating manual curation of text-	939
887	USA. Association for Computational Linguistics.	mined facts in biomedicine . <i>PLoS Computational</i>	940
888	Rodrigo Nogueira and Kyunghyun Cho. 2019. Pas-	<i>Biology</i> , 2(9):e118.	941
889	sage re-ranking with BERT . <i>arXiv preprint</i>	Shachar Rosenman, Alon Jacovi, and Yoav Goldberg.	942
890	<i>arXiv:1901.04085</i> .	2020. Exposing shallow heuristics of relation ex-	943
891	Abiola Obamuyide and Andreas Vlachos. 2018. Zero-	traction models with challenge data . In <i>Proceed-</i>	944
892	shot relation classification as textual entailment . In	<i>ings of the 2020 Conference on Empirical Methods</i>	945
893	<i>Proceedings of the First Workshop on Fact Extraction</i>	<i>in Natural Language Processing (EMNLP)</i> , pages	946
894	<i>and VERification (FEVER)</i> , pages 72–78, Brussels,	3702–3710, Online. Association for Computational	947
895	Belgium. Association for Computational Linguistics.	Linguistics.	948
896	Mihir Parmar, Swaroop Mishra, Mor Geva, and Chitta	Patrick Ruch, Emilie Pasche, Pierre-André Michel, and	949
897	Baral. 2023. Don’t blame the annotator: Bias al-	Déborah Caucheteur. 2024. BiotXplorer: Navigat-	950
898	ready starts in the annotation instructions . In <i>Pro-</i>	ing evidence-based biotic interactions . <i>Biodiversity</i>	951
899	<i>ceedings of the 17th Conference of the European</i>	<i>Information Science and Standards</i> , 8:e135453.	952
900	<i>Chapter of the Association for Computational Lin-</i>	Oscar Sainz, Oier Lopez de Lacalle, Gorka Labaka,	953
901	<i>guistics</i> , pages 1779–1789, Dubrovnik, Croatia. As-	Ander Barrena, and Eneko Agirre. 2021. Label ver-	954
902	sociation for Computational Linguistics.	balization and entailment for effective zero and few-	955
903	Hao Peng, Tianyu Gao, Xu Han, Yankai Lin, Peng	shot relation extraction . In <i>Proceedings of the 2021</i>	956
904	Li, Zhiyuan Liu, Maosong Sun, and Jie Zhou. 2020.	<i>Conference on Empirical Methods in Natural Lan-</i>	957
905	Learning from context or names? an empirical study	<i>guage Processing</i> , pages 1199–1212, Online and	958
906	on neural relation extraction . In <i>Proceedings of the</i>	Punta Cana, Dominican Republic. Association for	959
907	<i>2020 Conference on Empirical Methods in Natural</i>	Computational Linguistics.	960
908	<i>Language Processing (EMNLP)</i> , pages 3661–3672,	Tal Schuster, Darsh Shah, Yun Jie Serene Yeo, Daniel	961
909	Online. Association for Computational Linguistics.	Roberto Filizzola Ortiz, Enrico Santus, and Regina	962
910	Jorrit H Poelen, James D Simons, and Chris J Mungall.	Barzilay. 2019. Towards debiasing fact verification	963
911	2014. Global biotic interactions: An open infrastruc-	models . In <i>Proceedings of the 2019 Conference on</i>	964
912	ture to share and analyze species-interaction datasets.	<i>Empirical Methods in Natural Language Processing</i>	965
913	<i>Ecological Informatics</i> , 24:148–159.	<i>and the 9th International Joint Conference on Natu-</i>	966
		<i>ral Language Processing (EMNLP-IJCNLP)</i> , pages	967
		3419–3425, Hong Kong, China. Association for Com-	968
		putational Linguistics.	969

970	Akshay Swaminathan, Ivan Lopez, William Wang,	Dublin, Ireland. Association for Computational Lin-	1029
971	Ujwal Srivastava, Edward Tran, Aaroehi Bhargava-	guistics.	1030
972	Shah, Janet Y. Wu, Alexander L. Ren, Kaitlin Caoili,		
973	Brandon Bui, Layth Alkhani, Susan Lee, Nathan	Yuhao Zhang, Victor Zhong, Danqi Chen, Gabor Angeli,	1031
974	Mohit, Noel Seo, Nicholas Macedo, Winson Cheng,	and Christopher D. Manning. 2017. Position-aware	1032
975	Charles Liu, Reena Thomas, Jonathan H. Chen,	attention and supervised data improve slot filling .	1033
976	and Olivier Gevaert. 2024. Selective prediction	In <i>Proceedings of the 2017 Conference on Empiri-</i>	1034
977	for extracting unstructured clinical data . <i>Journal</i>	<i>cal Methods in Natural Language Processing</i> , pages	1035
978	<i>of the American Medical Informatics Association</i> ,	35–45, Copenhagen, Denmark. Association for Com-	1036
979	31(1):188–197.	putational Linguistics.	1037
980	Bruno Taillé, Vincent Guigue, Geoffrey Scoutheeten,		
981	and Patrick Gallinari. 2020. Let’s stop incorrect	Wenxuan Zhou and Muhao Chen. 2022. An improved	1038
982	comparisons in end-to-end relation extraction! In	baseline for sentence-level relation extraction . In <i>Pro-</i>	1039
983	<i>Proceedings of the 2020 Conference on Empirical</i>	<i>ceedings of the 2nd Conference of the Asia-Pacific</i>	1040
984	<i>Methods in Natural Language Processing (EMNLP)</i> ,	<i>Chapter of the Association for Computational Lin-</i>	1041
985	pages 3689–3701. Association for Computational	<i>guistics and the 12th International Joint Conference</i>	1042
986	Linguistics.	<i>on Natural Language Processing (Volume 2: Short</i>	1043
987	James Thorne, Andreas Vlachos, Christos	<i>Papers)</i> , pages 161–168, Online only. Association for	1044
988	Christodoulopoulos, and Arpit Mittal. 2018.	Computational Linguistics.	1045
989	FEVER: a large-scale dataset for fact extraction and		
990	VERification. In <i>Proceedings of NAACL-HLT</i> , pages	Heng-Xing Zou, Xiaohao Yang, Thabassum H. Haja-	1046
991	809–819.	maideen, Olivia J. Stein, Roxanne S. Beltran, Ben-	1047
992	David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu	jamin G. Freeman, Mark Lindquist, Eliot T. Miller,	1048
993	Wang, Madeleine van Zuylen, Arman Cohan, and	Summer Mengarelli, Charlotte M. Probst, Fernanda S.	1049
994	Hannaneh Hajishirzi. 2020. Fact or fiction: Verifying	Valdovinos, Derek Van Berkel, Phoebe L. Zarnetske,	1050
995	scientific claims . In <i>Proceedings of the 2020 Con-</i>	Brian C. Weeks, and Kai Zhu. 2026. Large language	1051
996	<i>ference on Empirical Methods in Natural Language</i>	models unlock the ecology of species interactions .	1052
997	<i>Processing (EMNLP)</i> , pages 7534–7550, Online. As-	<i>bioRxiv</i> . Preprint, posted 10 February 2026.	1053
998	sociation for Computational Linguistics.		
999	Peter West, Chandra Bhagavatula, Jack Hessel, Jena D.	A Teacher prompts	1054
1000	Hwang, Liwei Jiang, Ronan Le Bras, Ximing Lu,	The three questions put to the 32B teacher and the	1055
1001	Sean Welleck, and Yejin Choi. 2022. Symbolic	other zero-shot models (§4.4, Appendix E), verba-	1056
1002	knowledge distillation: from general language mod-	tim, at temperature 0 with thinking disabled; the	1057
1003	els to commonsense models . In <i>Proceedings of the</i>	score is the probability of YES among the YES/NO	1058
1004	<i>2022 Conference of the North American Chapter of</i>	alternatives of the first answer token. The triple	1059
1005	<i>the Association for Computational Linguistics: Hu-</i>	question is the prompt that labelled the training cor-	1060
1006	<i>man Language Technologies</i> , pages 4602–4625, Seat-	pus. On BioRED the sentence and pair questions	1061
1007	tle, United States. Association for Computational	name biomedical entities and relation types in place	1062
1008	Linguistics.	of organisms and biotic interactions, and there the	1063
1009	Michael Wiegand, Josef Ruppenhofer, and Thomas	pair question asks only whether the sentence states	1064
1010	Kleinbauer. 2019. Detection of abusive language:	a relation between the two entities beyond a mere	1065
1011	the problem of biased datasets . In <i>Proceedings of the</i>	co-mention (their exact text is released with the	1066
1012	<i>2019 Conference of the North American Chapter of</i>	code). Lines are wrapped for the page.	1067
1013	<i>the Association for Computational Linguistics: Hu-</i>		
1014	<i>man Language Technologies, Volume 1 (Long and</i>	Sentence question.	1068
1015	<i>Short Papers)</i> , pages 602–608, Minneapolis, Min-	You check sentences retrieved from the	1069
1016	nesota. Association for Computational Linguistics.	scientific literature for a	1070
1017	Shanchuan Wu and Yifan He. 2019. Enriching pre-	biotic-interaction database.	1071
1018	trained language model with entity information for		1072
1019	relation classification . In <i>Proceedings of the 28th</i>	Sentence: <passage>	1073
1020	<i>ACM International Conference on Information and</i>		1074
1021	<i>Knowledge Management (CIKM)</i> , pages 2361–2364,	Answer YES only if the sentence asserts	1075
1022	Beijing, China. Association for Computing Machin-	a direct biological interaction between	1076
1023	ery.	organisms (for example predation,	1077
1024	Michihiro Yasunaga, Jure Leskovec, and Percy Liang.	parasitism, infection, herbivory,	1078
1025	2022. LinkBERT: Pretraining language models with	pollination, mutualism, symbiosis or	1079
1026	document links. In <i>Proceedings of the 60th Annual</i>	disease transmission), and the	1080
1027	<i>Meeting of the Association for Computational Lin-</i>	interaction is not negated.	1081
1028	<i>guistics (Volume 1: Long Papers)</i> , pages 8003–8016,	Reply with exactly one word: YES or NO.	1082
			1083

Pair question.

You verify candidate biotic-interaction pairs extracted from scientific literature.

Sentence: <passage>

Candidate pair:

Entity 1: "<taxon 1>"

Entity 2: "<taxon 2>"

Answer YES only if ALL of the following hold:

1. Both entities are genuinely the organisms referred to by those surface strings in this sentence (not a gene, protein, chemical, author name, or a different species).
2. The sentence asserts a direct biological interaction between these two organisms (not merely a co-mention, and not an interaction each has with some third organism), and the interaction is not negated.

Reply with exactly one word: YES or NO.

Triple question (the labelling prompt).

You verify candidate biotic-interaction triples extracted from scientific literature.

Sentence: <passage>

Candidate triple:

Entity 1: "<taxon 1>"

Relation: "<relation>"

Entity 2: "<taxon 2>"

Answer YES only if ALL of the following hold:

1. Both entities are genuinely the organisms referred to by those surface strings in this sentence (not a gene, protein, chemical, author name, or a different species).
2. The sentence asserts a direct biological interaction between these two organisms (not merely a co-mention, and not an interaction each has with some third organism).
3. The stated relation is the correct type of interaction between them, in the stated direction, and is not negated.

Reply with exactly one word: YES or NO.

B Per-seed results

Table 2 gives the per-seed AUPRC behind the TAXAPAIRS means of Table 1.

C Values behind Figures 2 and 3

Table 3 gives the operating curves of Figure 2 at six thresholds, and Table 4 the AUPRC values of

seed	AUPRC			F1, $\tau = 0.5$		
	sent.	pair	triple	sent.	pair	triple
1	.871	.946	.948	.634	.804	.859
2	.868	.946	.947	.708	.772	.864
3	.860	.935	.939	.670	.802	.852
mean	.867	.942	.945	.671	.793	.858
sd	.006	.007	.005	.037	.018	.006

Table 2: Per-seed figures on the 437 TAXAPAIRS rows, order-free scoring. The body reports F1 of three-checkpoint ensembles at block-held-out thresholds; a fixed $\tau = 0.5$ is shown here only for the spread across seeds, not to compare the models, because the three models are calibrated differently, and a single shared threshold would favour one of them arbitrarily.

τ	sentence					pair				triple			
	P	R	F1	Acc.		P	R	F1	Acc.	P	R	F1	Acc.
0.01	.709	.920	.801	.746		.761	.976	.855	.737	.740	.940	.828	.730
0.04	.782	.817	.799	.600		.815	.948	.877	.668	.811	.924	.864	.654
0.10	.819	.737	.776	.517		.854	.908	.880	.611	.845	.912	.877	.620
0.25	.861	.693	.768	.462		.911	.853	.881	.538	.903	.849	.875	.540
0.50	.886	.526	.660	.341		.940	.689	.795	.421	.942	.777	.852	.474
0.90	.930	.159	.272	.098		.981	.422	.591	.247	.982	.637	.773	.373

Table 3: Values behind Figure 2: the three models' operating curves on all 437 rows, three-checkpoint ensembles; Acc. is the share of the 437 candidates the model accepts at τ . Rows show the shape; the reported operating points are those of Table 1.

Figure 3, with AUPRC by source block.

D BioRED in full

Table 5 gives the BioRED comparison of §4.2 in full: AUPRC by the number of concepts the sentence names (§4.3), the entity-marker variant, and the zero-shot teacher on a 3,000-candidate sample (§4.4).

E Zero-shot scaling

Table 6 gives the zero-shot comparison of Figure 4 in full, with the triple question added. Each model's score is $P(\text{YES})$ from its first answer token, and the triple question is the teacher's labelling prompt, asked in both argument orders and maxed like the trained triple-conditioned model. At block-held-out thresholds the 32B teacher's precision, recall and F1 are 0.654, 0.948 and 0.774 asked the sentence question, 0.919, 0.865 and 0.891 asked the pair question, and 0.878, 0.892 and 0.885 asked the triple question.

At 0.6B the three questions sit close to the base rate (0.66–0.68 against 0.574; 0.52 against 0.476): at that size the model barely does the task, whatever it is asked. From 1.7B on, the pair question

Candidates	<i>n</i>	Pos.	Sentence	Pair	Triple
passage names ≤ 2 taxa	161	0.590	0.896	0.939	0.930
passage names ≥ 3 taxa	276	0.565	0.857	0.955	0.963
both arguments are taxa, ≤ 2	33	0.636	0.958	0.984	0.976
both arguments are taxa, ≥ 3	227	0.577	0.891	0.975	0.982
an argument is not a taxon, ≤ 2	128	0.578	0.876	0.925	0.915
an argument is not a taxon, ≥ 3	49	0.510	0.628	0.808	0.839
TEST299	299	0.548	0.835	0.944	0.951
BIOTX100	97	0.784	0.964	0.982	0.973
REJECT50	41	0.268	0.551	0.575	0.675

Table 4: Exact values of Figure 3 (TAXAPAIRS rows) and AUPRC by source block; all values are AUPRC of three-checkpoint ensembles, and Pos. is the share of positives in the row, the AUPRC of a random ranking. Taxa are counted as distinct mentions found in the passage by TaxoNERD (Le Guillarme and Thuiller, 2022) (en_ner_eco_biobert); an argument “is not a taxon” when TaxoNERD does not recognise it in the passage. TEST299 passages name 5.1 taxa on average, BIOTX100 2.6 and REJECT50 1.7.

is ahead at each size on both benchmarks: by +0.039 AUPRC at 1.7B and by +0.09 to +0.20 from 4B on (+0.11, then +0.20 to +0.30, on BioRED). The sentence question gains little with size (0.76–0.82 on TAXAPAIRS from 4B on; 0.52–0.57 on BioRED throughout, near its base rate), so scale mostly improves the answer to the right question. The lead is not monotonic in size among the Qwen3 models, and it is largest for the 122B model: +0.197 on TAXAPAIRS and +0.301 on BioRED, where the 32B model’s is +0.246. Asked the triple question, every Qwen3 model from 1.7B to 30B-A3B scores above its answer to the pair question on TAXAPAIRS, while the 0.6B, 32B and 122B models score level with it. Asked about the pair, the 122B model ranks the 437 candidates better than any trained model (0.985; the recommended model 0.970, the pair-conditioned student 0.942 per-seed mean, 0.947 as an ensemble), which suggests that a stronger teacher would raise the students further, whereas soft labels and larger encoders raise them little (Appendices F and H); asked the sentence question, it scores 0.788.

F Soft-label distillation

The students learn from the teacher’s verdicts. We re-queried the teacher for $P(\text{YES})$ on all 48,338 training rows (same prompt and inputs; its verdicts agree with the corpus labels on 99.7% of rows, and 24% of the probabilities lie between 0.05 and 0.95) and trained the three models again against these soft targets with a soft cross-entropy, selecting checkpoints on the same hard-labelled devel-

Model	AUPRC	F1	AUPRC, sentences naming		EP F1 (%)
			2 concepts	≥ 3	
Sentence-level	0.738 $\pm$.004	0.718	0.909	0.727	57.5
Pair-conditioned	0.843 $\pm$.004	0.817	0.902	0.847	67.1
+ entity markers	0.840 $\pm$.003	0.815	0.907	0.844	66.5
<i>Zero-shot 32B teacher, 3,000-candidate sample</i>					
Sentence question	0.566	0.662	0.921	0.555	n/a
Pair question	0.813	0.766	0.958	0.799	n/a

Table 5: BioRED, BioCreative VIII test set: 16,606 sentence-level candidates from 400 abstracts, 0.463 positive, 538 from sentences naming two concepts. Headline rows repeat Table 1. AUPRC is the per-seed mean $\pm$ standard deviation over three seeds (columns by concept count: three-checkpoint ensembles); F1 is at the pre-specified threshold (mean of the checkpoints’ development thresholds), the teacher’s at its greedy answer. EP F1 is BioRED’s document-level entity-pair F1, a pair scoring the maximum over the sentences that mention it. On the teacher’s sample the trained models’ three-checkpoint ensembles score 0.757 and 0.860. Pair-conditioned against sentence-level: 2,642 items fixed, 1,064 broken (McNemar $p = 5.9 \times 10^{-148}$).

opment split. It is the only distillation variant we tried, scored once (Table 7). Every model gains a little AUPRC (+0.005 to +0.007), the triple-conditioned model on every seed and with 0.021 F1 ($p = 0.03$ before correcting for three comparisons); against the pair-conditioned model it is marginally ahead (0.951 against 0.947; 25 items fixed, 21 broken, $p = 0.66$). The comparison is unchanged: the pair query lifts AUPRC from 0.873 to 0.947 (72 fixed, 28 broken, $p = 1.7 \times 10^{-5}$), by +0.095 on passages naming three or more taxa and +0.036 on the rest. Probabilities close about a quarter of the pair-conditioned model’s distance to its teacher’s 0.964 (0.016 AUPRC, against 0.021 with verdicts). The body keeps the verdict-trained models, on which every other analysis was run.

G Query ablation

Degrading the trained triple-conditioned model’s query at inference, with weights and thresholds unchanged, shows which part of the query it uses (Table 8; §4.5). Deleting both taxa collapses ranking from 0.948 to 0.691 against a base rate of 0.574, and emptying the query entirely gives 0.649; replacing or shuffling the relation, pair kept, moves ranking far less (0.906, 0.930): the model leans on the pair, which is what the pair-conditioned model already showed.

The empty-query variant is not the sentence-level model: deprived of the query it was trained with, it does far worse (0.649) than a model trained

Qwen3	TAXAPAIRS, 437				BioRED, 3,000		
	sent.	pair	triple	Δ	sent.	pair	Δ
0.6B	0.676	0.658	0.663	-0.018	0.522	0.523	+0.000
1.7B	0.769	0.808	0.844	+0.039	0.533	0.648	+0.115
4B	0.761	0.885	0.921	+0.124	0.546	0.745	+0.199
8B	0.819	0.909	0.947	+0.090	0.524	0.800	+0.275
14B	0.818	0.929	0.959	+0.112	0.555	0.805	+0.250
30B-A3B	0.797	0.927	0.956	+0.130	0.571	0.792	+0.221
32B (teacher)	0.804	0.960	0.964	+0.156	0.566	0.813	+0.246
Qwen3.5-122B-A10B	0.788	0.985	0.984	+0.197	0.519	0.820	+0.301
trained, same rows	0.871	0.947	0.948	+0.076	0.757	0.860	+0.103

Table 6: AUPRC of zero-shot Qwen3 (Qwen Team, 2025) by size, asked the three questions of Appendix A (4-bit builds served locally, thinking disabled; the triple question in both argument orders, maxed). 30B-A3B is a mixture of experts with 3B active parameters; Qwen3.5-122B-A10B (Qwen Team, 2026), a later release, is a mixture of experts with 8 of 256 experts active per token, run partly on the CPU. Base rates 0.574 and 0.476. Last row: the trained three-checkpoint ensembles on the same rows. Δ : pair minus sentence question (last row: pair-conditioned minus sentence-level model).

without one (0.867); hence three trained models.

H Encoders

Besides the recipe of §3.2 we tried seven others on development data only: five epochs at learning rates 1, 2 and 3×10^{-5} , BiomedBERT-large at 1 and 2×10^{-5} , and BioLinkBERT-base and -large, pretrained on PubMed with citation links (Yasunaga et al., 2022). The three that beat the base recipe by 0.01 development AUPRC or more on both splits (seed 1, pair-conditioned model) were scored once on the test sets: BiomedBERT-large at 2×10^{-5} (Appendix P), and BioLinkBERT in its base size under the base recipe and in its large size (below). BioLinkBERT-base’s development AUPRC is 0.910 for the BioRED pair-conditioned model over three seeds, against 0.892 for the base encoder and 0.905 for BiomedBERT-large, and 0.852 for the biodiversity pair-conditioned model on seed 1, then 0.843 and 0.833 on seeds 2 and 3. Its BioRED pair-conditioned model gains on every seed (AUPRC 0.853–0.858 against 0.839–0.847; EP F1 67.7–68.0 against 66.8–67.5), the best sentence-scope AUPRC we have on BioRED (Table 9). On TAXAPAIRS its pair-conditioned model does not gain (0.939 ± 0.004 against 0.942 ± 0.007 ; F1 0.860 against 0.882) while its sentence-level model does (0.888, F1 0.816 against 0.790), so the formulation gap narrows to +0.051 AUPRC (53 items fixed, 37 broken, $p = 0.11$), still concentrated on passages naming three or more taxa

Model	Target	AUPRC	F1	AUPRC, taxa	
				≤ 2	≥ 3
Sentence	verdict	$0.867 \pm .006$	0.790	0.896	0.857
	$P(\text{YES})$	$0.873 \pm .007$	0.799	0.908	0.862
Pair	verdict	$0.942 \pm .007$	0.882	0.939	0.955
	$P(\text{YES})$	$0.947 \pm .003$	0.883	0.944	0.957
Triple	verdict	$0.945 \pm .005$	0.867	0.930	0.963
	$P(\text{YES})$	$0.951 \pm .003$	0.888	0.936	0.965

Table 7: The sentence-level, pair-conditioned and triple-conditioned models trained on the teacher’s verdicts (the body’s models) and on its probabilities, recipe and seeds unchanged. AUPRC is the per-seed mean $\pm$ standard deviation; F1 (block-held-out thresholds) and AUPRC by the number of taxa the passage names are those of three-checkpoint ensembles, as in Tables 1 and 4. Per-seed AUPRC with soft targets: .866/.879/.875, .951/.946/.945 and .952/.952/.948. Against the verdict-trained model of the same format, the soft-target model fixes 13 items and breaks 7 (sentence-level), 13 and 14 (pair-conditioned), 19 and 7 (triple-conditioned; McNemar $p = 0.26, 1.0, 0.03$).

Query given to the model	P	R	F1	AUPRC
Retrieved triple	0.842	0.892	0.867	0.948
Generic relation, pair kept	0.918	0.673	0.777	0.906
Relation shuffled across items	0.900	0.721	0.801	0.930
Relation only, no taxa	0.833	0.020	0.039	0.691
Empty query	1.000	0.036	0.069	0.649

Table 8: Degrading segment A of the trained triple-conditioned model at inference, weights and its own block-held-out thresholds unchanged; AUPRC of the three-checkpoint ensemble. Base rate 0.574. Per-seed standard deviations on AUPRC are ≤ 0.032 .

(+0.073 against +0.009). On BioRED it widens, to +0.113.

BioLinkBERT-large, at 2×10^{-5} for three epochs like BiomedBERT-large, passed the same screen (development AUPRC 0.907 on BioRED and 0.869 on the biodiversity split, seed 1; three-seed means 0.903 and 0.862). Its pair-conditioned models gain a little on both benchmarks: on BioRED, AUPRC 0.848 ± 0.009 and EP F1 68.2 ± 0.7 (seeds 67.7–68.9, each above every base seed); on TAXAPAIRS, 0.949 ± 0.003 and F1 0.896. Neither gain is significant: against the base encoder’s pair-conditioned model it fixes 652 BioRED items and breaks 653 ($p = 1.0$), and on TAXAPAIRS 28 and 22 ($p = 0.48$). Its other models, trained afterwards and scored once, complete the comparison: the sentence-level models reach 0.881 on TAXAPAIRS and 0.737 on BioRED (EP F1 56.1), the triple-conditioned model 0.951, and the pair query again wins, by +0.068 on TAXAPAIRS (74

Encoder	BioRED, BC8 test			TAXAPAIRS, 437			
	sent.	pair	EP F1 (%)	sent.	pair	triple	pair F1
BiomedBERT-base	0.738	0.843	57.5/67.1	0.867	0.942	0.945	0.882
BiomedBERT-large	0.741	0.843	57.6/67.4	n/a	n/a	n/a	n/a
BioLinkBERT-base	0.743	0.855	57.9/67.9	0.888	0.939	0.949	0.860
BioLinkBERT-large	0.737	0.848	56.1/ 68.2	0.881	0.949	0.951	0.896

Table 9: The three formulations under the encoders that were scored on test, recipe otherwise that of §3.2: AUPRC and EP F1 (sentence-level/pair-conditioned) as per-seed means over three seeds, and block-held-out F1 of three-checkpoint ensembles. BiomedBERT-large was not trained on the biodiversity arms (Appendix P). Against the base encoder’s model of the same format, BioLinkBERT’s pair-conditioned model fixes 621 BioRED items and breaks 537 (McNemar $p = 0.015$) and on TAXAPAIRS 19 and 32 ($p = 0.093$); its sentence-level model on TAXAPAIRS, 42 and 20 ($p = 0.0077$). No BioLinkBERT-large model differs significantly from the base encoder’s on TAXAPAIRS ($p \geq 0.23$).

items fixed, 28 broken, $p = 8 \times 10^{-6}$; +0.090 on passages naming three or more taxa, +0.026 on the rest) and by +0.111 on BioRED. We keep the base encoder: BioLinkBERT-base does not help the pair-conditioned model on TAXAPAIRS, and BioLinkBERT-large’s gains are not significant on either benchmark.

I Recall on rejected candidates

REJECT50 asks whether candidates an earlier filter rejected contain recoverable positives, and it is too small to rank the models: eleven positives, so one item moves recall by nine points. At block-held-out thresholds the sentence-level model recovers 9 of the 11 by re-admitting 13 of the 30 correct rejections (F1 0.545), the pair-conditioned model recovers 6 while re-admitting 5 (0.545), and the triple-conditioned model recovers 8 while re-admitting 3 (0.727). We read only the qualitative point: the discarded candidates do contain recoverable interactions, and a model that reads the candidate finds them with fewer wrong re-admissions than one that reads only the passage.

J Error examples

The examples illustrate the mechanisms of §4.6, one each (Table 10).

K Entity disambiguation and direction

Taxonomy gate. The natural remedy for entity errors is to gate candidates on taxonomic validity. Against the Open Tree Taxonomy (Rees and

Cranston, 2017) such a gate catches 1 of the 32 unsupported BIOTX100 candidates and discards 5 of the 65 supported ones. This is because *Illiger*, *Dikarya*, *copia* and *sulla*, the surnames, clades, transposon families and vernacular names behind the entity failures, all resolve in the taxonomy. The failure is therefore one of disambiguation, of the kind species-name normalisers address (Gerner et al., 2010), which a validity gate cannot detect; and the gate’s casualties are viruses, which are legitimate partners in this domain. The *authority* and *same_organism* rules (Appendix L) target the disambiguation directly and are the better instrument.

Direction labels. The benchmark’s expert grading marks reversed triples as correct; we verified two by hand, (*Breinlia*, host of, *Rattus*) and (*Demi-dospermus*, parasitized by, *A. inermis*). No directional number on the benchmark would therefore measure the model rather than the convention, which is why direction is scored on its own annotation (Appendix M). The supervision itself is also uneven: among the 5,591 reversed training rows, the rate at which the teacher rejects a reversal is governed by the relation’s surface string. Direction-encoding participles and prepositional forms are rejected at 0.51–0.70 (*affecting* 0.70, *affected by* 0.61, *parasite of* 0.57), bare relational nouns at 0.04–0.10 (*host of* 0.08, *pollinators* 0.04). Whether a training label encodes direction therefore depends on which string retrieval matched.

L Candidate rules

Several error mechanisms of §4.6 are questions about the candidate rather than the passage (whether two strings name one organism, whether a word is a surname, whether *relative to* is an interaction), and a classifier handles them poorly. We answer them with eight deterministic rejection rules, applied to the candidate before the verifier decides, each targeting one of those mechanisms. *same_organism* (apposition of a common and a scientific name, or identical forms), *own_clade* (an adjacent clade annotation such as “(Acari: Prostigmata)”), and *authority* (a taxonomic author parsed as a taxon) detect the same-organism and entity classes. *non_biotic_relation* rejects phylogenetic, comparative and biogeographic relation terms; *negated* rejects the two negation forms whose scope provably contains one of the pair (“non-*s* hosts”, “no infections with ... *s*”); *not_an_organism* rejects organs and syn-

Mechanism	Candidate triple	Passage (abridged)
Pair-binding	(<i>T. gondii</i> , parasite of, swine)	“The entero-pathogenic <i>Cystoisospora suis</i> , a coccidian par- asite of swine and close relative to <i>Toxoplasma gondii</i> ...”
Co-listing	(katydid, prey, beetles)	“... remains were from insects ... (e.g. beetles , dragonflies, cockroaches, and female katydids), indicating that eaves-dropping on prey signals is not the only prey-finding strategy used by this <i>bat</i> .”
Negation	(wheat, hosts, <i>F. acuminatum</i>)	“... <i>F. acuminatum</i> in New Zealand was found only on non-wheat hosts .”
Same organism	(<i>P. chinensis</i> , pollinators, Chinese knotweed)	“... potential pollinators of Chinese knotweed (<i>Persicaria chinensis</i>) .”
Entity	(Siberia , migrating, birds)	“... a portion of birds migrating from Siberia have common winter nesting areas.”

Table 10: One example of each mechanism among the pair-conditioned model’s false positives. The relation’s true argument is italicised where it is not the queried taxon; in the entity case a place name was matched as a taxon.

System	P	R	F1	fixed	broken
Pair-conditioned	0.869	0.896	0.882	n/a	n/a
+ 7 rules	0.878	0.888	0.883	3	2
+ 8 rules	0.887	0.876	0.882	6	5
Triple-conditioned	0.842	0.892	0.867	n/a	n/a
+ 7 rules	0.860	0.880	0.870	6	3
+ 8 rules	0.883	0.869	0.876	13	6

Table 11: Deterministic candidate rules applied before the model’s decision, all 437 rows, three-checkpoint ensembles at block-held-out thresholds. “Fixed” and “broken” are relative to the same model without rules. McNemar: $p = 1.00$ and 1.00 on the pair-conditioned model, 0.50 and 0.17 on the triple-conditioned model.

dromes; *pathogen_modifier* rejects an adjective that only ever occurs inside a pathogen’s name; and *co_listed* rejects two taxa that are conjuncts of one dependency-parsed list not governed by an interaction noun (“interactions between *A* and *B*” is kept).

Table 11 gives their effect on both models. On the pair-conditioned model they raise precision by 0.009 to 0.018 and lower recall by 0.008 to 0.020, F1 unchanged ($p = 1.00$): of its 34 false positives they catch 6, because a model told which pair it is judging already rejects most of what they target. On the triple-conditioned model they do more (precision $0.842 \rightarrow 0.883$, $p = 0.17$). Both are upper estimates, since half the rules were chosen with this benchmark’s errors in view (below); their main practical benefit is transparency, because each rejection names its rule.

Validation. No rule has a parameter fitted to evaluation labels, but rules can still be fitted by choice, and four of these (*non_biotic_relation*, *negated*, *not_an_organism* and *pathogen_modifier*) were written after reading the false positives of §4.6. To

Expert-graded label	<i>n</i>	answered	correct
forward or reverse	84	62 (74%)	55 of 62 (0.887)
undecidable (“?”)	12	n/a	9 left UNCERTAIN
bidirectional	1	n/a	1 BIDIRECTIONAL

Table 12: The recommended model’s four-way direction output against 97 expert-graded direction labels. “Answered” counts FORWARD or REVERSE; the rest are UNCERTAIN. The abstention level (0.60) was chosen by the expert after seeing this set’s coverage curve, so the figures are optimistic.

keep their selection from simply memorising the benchmark, every rule had to pass a criterion fixed before it was evaluated: on the 42,747 training rows in retrieved argument order, at least 85% of the candidates it rejects must be teacher-negative, matching the teacher’s own accuracy. Two first drafts failed it and were narrowed on training data alone (a general negation cue fired on “without affecting the host”, and a role-noun list rejected “viruses” and “fungi”, which are legitimate coarse taxa), and the final rules reject teacher-negatives at 0.88 to 1.00 (*co_listed*, which needs a dependency parse, on a fixed 6,000-row sample). The rules were then frozen, recorded by hash, and evaluated once.

M The recommended configuration

The model we recommend for use (§5) lies outside the controlled comparison. It is one 110M-parameter model with two heads, and it differs from the pair-conditioned model of §3.2 in three respects adopted during development: it gives the two taxa in alphabetical order, so the input is identical whichever way round retrieval presents the pair; it marks them in place in the passage (@...@, #...#) rather than only naming them in segment A;

and its training labels are the teacher’s pair-level judgments. Its interaction threshold, 0.5, was set before evaluation and never tuned. On the 437 candidates it reaches AUPRC 0.970 and F1 0.915 (precision 0.873, recall 0.960), and with the candidate rules of Appendix L it reaches precision 0.888, recall 0.948 and F1 0.917 (5 items fixed, 3 broken, $p = 0.72$).² It scores 32.3 candidates per second on eight CPU threads, measured with both heads. Escalating the 30% of candidates nearest its threshold to a zero-shot Qwen3.5-122B-A10B raises its precision with the rules from 0.888 to 0.962 at recall 0.912 (Appendix N).

Direction head. The second head reports the interaction’s *direction* in four categories. FORWARD and REVERSE say which taxon acts on the other; they come from a direction score that is exactly antisymmetric in the two marked taxa, and are reported only when it is confident, otherwise the answer is UNCERTAIN. BIDIRECTIONAL is assigned when the relation is mutual by definition (mutualism, symbiosis, competition, co-infection, *interacts with*), as listed in a relation-polarity lexicon derived from the Relation Ontology; for those, neither taxon is the subject and the model is not asked. The benchmark’s expert grading scores reversed triples as correct (Appendix K), so direction is evaluated on 97 separately expert-graded candidates (Table 12): the model answers 74% of the directional ones, correctly on 0.887 of those, an optimistic figure since the abstention level was chosen on that set.

N Escalating uncertain candidates to the teacher

The recommended model is fast and recall-oriented, and the zero-shot teacher, asked about the pair, is precise but slow (Appendix E). Table 13 combines them: the model decides every candidate, and those it is least sure of (a fixed fraction, nearest its threshold) are re-decided by a zero-shot LLM. Escalating 30% to Qwen3.5-122B-A10B raises precision from 0.888 to 0.962 at recall 0.912, removing 21 of the 30 false positives; the model alone reaches that precision only at $\tau = 0.99$, where recall falls to 0.745. Against the model alone the decisions are not significantly better on these 437 rows (25 items fixed, 13 broken, McNemar $p = 0.07$; with Qwen3-32B,

²The *co_listed* rule is omitted from this configuration: on this model’s accept set it removes one false positive and three true ones.

Band	LLM calls	Qwen3-32B			Qwen3.5-122B-A10B		
		P	R	F1	P	R	F1
0%	0	0.888	0.948	0.917	0.888	0.948	0.917
10%	44	0.908	0.940	0.924	0.912	0.944	0.928
20%	87	0.920	0.920	0.920	0.932	0.924	0.928
30%	131	0.950	0.912	0.931	0.962	0.912	0.937
50%	218	0.964	0.849	0.903	0.981	0.841	0.906
100%	437	0.960	0.773	0.857	0.984	0.757	0.856

Table 13: The recommended model with its candidate rules keeps its own decision except on the band of candidates whose score lies nearest its threshold, which take a zero-shot LLM’s answer to the pair question (Qwen3-32B, the teacher, or Qwen3.5-122B-A10B); 437 rows. A rule rejection is final. Band 0% is the recommended model alone, 100% the LLM alone (with the rules).

22 and 13, $p = 0.18$): the gain lies in the operating point, with more recall at a given precision, while accuracy at the model’s own threshold does not improve significantly. We report the band at six settings and do not choose one. Its cost is set by the teacher’s speed: escalating 30% of the 175,588-candidate pool means about 53,000 calls, under a day on one A100 for Qwen3.5-122B-A10B (1.4 s per call with part of it on the CPU) and less for the 32B.

O Artifacts and compute

The training recipe of §3.2 is AdamW at learning rate 2×10^{-5} and weight decay 0.01, a linear schedule with 10% warm-up, batch size 32, inputs of at most 256 wordpieces, three epochs and gradient clipping at 1.0; Appendix H lists the alternatives we screened. The encoders are BiomedBERT-base (microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext) and -large (MIT licence) and BioLinkBERT-base and -large (Apache 2.0); the zero-shot models are Qwen3 0.6B-32B and Qwen3.5-122B-A10B (Qwen Team, 2026) (Apache 2.0), run locally through Ollama; taxon mentions are counted with TaxoNERD (MIT). BioRED and its BioREDirect directionality release are public NCBI resources, used as distributed. Everything ran on one NVIDIA A100 (80 GB): an encoder run takes between two and fifteen minutes, and the whole study took about 50 GPU-hours, most of them LLM inference for labelling the corpus and for the zero-shot rows.

Intervention	Benefit	Drawback
Taxon-validity gate	1 of 32 unsupported caught	−5/65 supported; rejects viruses
Self-distillation [†]	none	−0.001 P, −0.008 F1
6-layer compression [†]	1.9× CPU throughput	−0.088 precision
Corpus signal-matching [†]	none	−0.059 AUPRC
Rule-family training data [†]	none	AUPRC +0.005 to −0.037
BiomedBERT-large	+0.013 dev AUPRC (BioRED pair)	none on test; 3× parameters

Table 14: Six plausible design choices whose drawbacks outweighed their benefits in our measurements. [†]Measured under an earlier version of the evaluation; deltas are directions only (see text). The last row compares per-seed means.

is unchanged (+0.103 against +0.105 AUPRC). It was not trained on all biodiversity arms, since its BioRED test result could not change the choice. For the whole-abstract BioRED model of §4.2 we also tried, on development data only (seed 1), eight epochs instead of three, alone and with each concept’s BioRED type in the query: development AUPRC 0.787 and 0.791 against 0.786, inside the spread of the three-epoch seeds (0.765–0.786), so neither was trained further or scored on the test set.

P Negative results

Table 14 collects interventions that did not help, because each is a plausible first thing to try. Self-distillation at equal capacity is null, suggesting the teacher’s knowledge is already saturated in the student. Compressing to six layers nearly doubles CPU throughput, which this setting does not need, and lowers precision by 0.088. Reweighting the corpus toward the signal distribution of the retrieval pool lowered AUPRC by 0.059. The fifth row is the most informative. Before the candidate rules existed, we turned the same mechanisms into training data (synthetic negatives for the same-organism, own-clade and authority classes, and in a second variant for co-listing) and re-trained. The first variant left AUPRC unchanged within seed noise and the second lowered it by 0.037. For properties of the candidate itself, rules worked better than extra training data. Deltas in rows two to five were measured under an earlier version of the evaluation and are given as directions. The last row is one of the three recipes of §4.5 that passed the development screen (the others are in Appendix H): BiomedBERT-large raises the BioRED pair-conditioned model’s development AUPRC from 0.892 to 0.905 (three-seed means), but on the BC8 test set the pair-conditioned model scores 0.843 ± 0.005 against 0.843 ± 0.004 , F1 0.814 against 0.817, EP F1 67.4 ± 0.4 against 67.1 ± 0.3 , and the sentence-level model 0.741 against 0.738; the pair query’s gain